

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November - December 2012

CL 213 Building Construction

Date: 01.12.2012, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Define following terms: [3]
(i) Baluster (ii) Newel post (iii) Scotia
- (b) Enlist various types of fixtures and fastenings used in doors & windows. Draw neat sketches of any one of each (both for fixtures and fastenings). [4]
- Q - 2 (a) Explain dormer and gable window with sketch. Also suggest the appropriate location and uses of both. [6]
- (b) Draw a neat sketch of hollow core flush door with labeling. [4]
- (c) Sketch a bifurcated stair. [4]

OR

- Q - 2 (a) Explain collapsible steel door with sketch. Discuss construction suitability of the same. [5]
- (b) Discuss about casement window with figure. [5]
- (c) Prepare a neat sketch of dog-legged staircase. [4]
- Q - 3 (a) List out types of roof covering. Enlist and discuss factors affecting for selection of roof covering. [7]
- (b) (i) List out factors affecting selection of flooring material. [3]
(ii) Draw a sketch of jack arch type flooring showing all components. [4]

OR

- Q - 3 (a) Enlist various pitched or sloping roof. Explain any one type with labeled sketch. [7]
- (b) (i) Show the components of a floor in neat figure. [3]
(ii) Draw a sketch of brick flooring showing all components. [4]

SECTION - II

- Q - 4** Design strip footing for a brick wall 30 cm thick and 3.2 m high above the ground level. The wall carries a superimposed load of 95 kN/m run. The soil has unit weight of 17 kN/m³, angle of repose of 30° and safe bearing capacity 150 kN/m². The footing have lime concrete base with unit weight of 20 kN/m³ and modulus of rupture 120kN/m². Assume unit weight of masonry is 19.5 kN/m³. [7]
- Q - 5 (a)** Draw a neat sketch of grillage foundation for steel stanchion. [4]
- (b)** Explain following stone masonry works with neat figures: [4]
- (i) Random rubble uncoursed & built to course
- (ii) Square rubble uncoursed & built to course
- (c)** Differentiate between following: [6]
- (i) Joint & bond (ii) Header & stretcher (iii) Lap & perpend
- (iv) Bat & closer (v) Junctions & quoins (vi) Acute & obtuse squint

OR

- Q - 5 (a)** Briefly explain seismic refraction and electrical resistivity methods of site exploration. [4]
- (b)** Draw elevation and plan of alternate courses of English bond for 1½ brick thick wall. [5]
- (c)** Discuss the joints used in stone masonry with neat figures. [5]
- Q - 6 (a)** Explain different forms of brick-stone composite masonry work. [2]
- (b)** (i) Enlist and discuss defects observed in plastering. [3]
- (ii) Explain the methods of pointing with neat sketches. [3]
- (c)** Explain following methods with neat diagrams: [6]
- (i) Pit method (ii) Mason's scaffolding (iii) Single flying shore

OR

- Q - 6 (a)** Write the rules for brick bonding. [2]
- (b)** (i) Discuss in detail defects observed in painting. [3]
- (ii) What do you understand by plastering & pointing? What are the objectives of plastering? [3]
- (c)** Write short notes on following methods: [6]
- (i) Pile method (ii) Box sheeting (iii) Double flying shore
